

1. Given the sorted data

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

27 observations

a) Mean & Median

$$\text{Mean} = \frac{\sum_{i=1}^n x_i}{n} = \frac{809}{27} = 29.96 \approx 30$$

$$\text{Median} = \frac{27+1}{2} = 14^{\text{th}} \text{ value} \quad \boxed{\text{Median} = 25}$$

b) Mode and Modality

• Modes = 25 and 35

= Bimodal (As only 2 values are repeating)

c) Midrange

$$\frac{\text{Max} + \text{Min}}{2} = \frac{13 + 70}{2} = \frac{83}{2} = 41.5$$

d)  $Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70$

$$Q_1 = 20$$

$$Q_3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70$$

$$Q_3 = 35$$

e) 5 number summary

13, 20, 25, 35, 70

f) Box plot:

13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36, 40,

45, 46, 52, 70

2. Ages

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

$$\text{Mean} = \frac{330}{10} = 33$$

15, 11, 8, 9, 5, 10, 0, 2, 23, 5

Mean deviation:

$$= \frac{15 + 11 + 8 + 9 + 5 + 10 + 2 + 23 + 5}{10}$$

$$= \frac{88}{10} = 8.8 = \text{MAD}$$





3. Suppose that the data for analysis includes the attribute age. The age values for the data are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data using a bin depth of 3. Illustrate your steps.

|       |              |   |              |
|-------|--------------|---|--------------|
| Bin 1 | - 13, 15, 16 | → | Mean = 14.67 |
| 2     | - 16, 19, 20 | → | = 18.33      |
| 3     | - 20, 21, 22 | → | = 21.00      |
| 4     | - 22, 25, 25 | → | = 24.00      |
| 5     | - 25, 25, 30 | → | = 26.67      |
| 6     | - 33, 33, 35 | → | = 33.67      |
| 7     | - 35, 35, 35 | → | = 35.00      |
| 8     | - 36, 40, 45 | → | = 40.33      |
| 9     | - 46, 52, 70 | → | = 56.00      |

Replace the mean for all three values in the Bin

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

- min-max normalization by setting min=0 and max=1
- Z-score normalization.

original value

Normalized

200

$$\frac{200-200}{800} = 0$$

300

$$\frac{300-200}{800} = \frac{1}{8}$$

400

$$= \frac{1}{4}$$

600

$$= \frac{1}{2}$$

1000

$$= 1$$

min-max normalization = 0, 0.125, 0.25, 0.5, 1

b)  $z = \frac{x - \mu}{\sigma}$

$$\mu = \frac{2500}{5} = 500$$

-x

x - μ

(x - μ)<sup>2</sup>

200

-300

90000

300

-200

40000

400

-100

10000

600

100

10000

1000

500

250000

$$\sum (x - \mu)^2 = 400000$$

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma = \sqrt{80000}$$

$$\sigma = \sqrt{8} \times 10^3$$

$$\sigma = 282.84$$



|    |      |    |      |
|----|------|----|------|
| 23 | 4.0  | 54 | 42.5 |
| 23 | 26.5 | 54 | 28.8 |
| 27 | 7.8  | 56 | 33.4 |
| 27 | 17.8 | 57 | 30.2 |
| 39 | 31.4 | 58 | 34.1 |
| 41 | 25.9 | 58 | 32.9 |
| 47 | 27.4 | 60 | 41.2 |
| 49 | 27.2 | 61 | 35.7 |
| 50 | 31.2 |    |      |

4

Age

$$\text{Mean} = 46.44$$

$$\text{Median} = 51$$

$$\text{SD} = 13.22$$

fat

$$\text{Mean} = 28.78$$

$$\text{Median} = 30.70$$

$$\text{S.D} = 9.25$$

b) Box plot:

23, 33, 51, 57, 61

6. Given sorted data

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

a) Equal frequency partitioning

No. of records = 12

bins = 3

Records per bin =  $\frac{12}{3} = 4$

| Bin | record           |
|-----|------------------|
| 1   | 5, 10, 11, 13    |
| 2   | 15, 35, 50, 55   |
| 3   | 72, 92, 204, 215 |

b) Equal width Partitioning

Min = 5    Max = 215

width =  $\frac{215-5}{3} = 70$

Intervals

|       |         |
|-------|---------|
| Bin 1 | 5-74    |
| 2     | 75-144  |
| 3     | 145-215 |

c) clustering:

| cluster (Bin) | Records            |
|---------------|--------------------|
| 1             | 5, 10, 11, 13, 15  |
| 2             | 35, 50, 55, 72, 92 |
| 3             | 204, 215           |